



LEONIDAS LIFTS



THE COMPLETE TRAINING PROGRAM

6-Day Push / Pull / Shoulders & Legs Split

Built on 18 Years of Natural Lifting

No Steroids. Just Work.

WITH RECOVERY SCIENCE ADDENDUM

◆ BACKED BY 56 PEER-REVIEWED REFERENCES ◆

LEONIDASLIFTS.COM

THIS IS NOT MEDICAL OR HEALTH ADVICE. FOR RESEARCH & EDUCATIONAL PURPOSES ONLY. CONSULT A PHYSICIAN BEFORE USE.

◆ PUSH DAY — CHEST & TRICEPS ◆

MONDAY & THURSDAY | PERFORMED 2x PER WEEK

EXERCISE	SETS	REP TARGET	NOTES
Bench Press (Progressive Overload)	9	80-100	3x15 light > 3x10 increasing > 1x8 struggle > 2x3-5 heavy
Cable Flys	3	30-45	Full stretch, controlled tempo
Light Dumbbell Fly Stretch	3	30-45	Deep stretch at bottom, light weight
Skullcrushers	4	100	Volume accumulation for tricep long head
Tricep Pulldown Trifecta (5 Cable Variations)	10	150-200	Rope / V-Bar / Straight / Reverse / Single — 2 sets each

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

◆ *Progressive overload on bench is the cornerstone. Track your weights. Beat the logbook every session.*

RESEARCH: WHY THESE EXERCISES

Bench Press: The flat bench press maximizes sternal and clavicular pectoralis major activation, making it the gold standard for chest development. EMG analysis shows flat bench produces the highest overall pec recruitment compared to incline variations (Rodriguez-Ridao et al., 2020). Non-uniform fiber recruitment means both flat and incline angles are valuable, but flat bench remains the primary mass builder (Cabral et al., 2021). The 9-set pyramid structure ensures progressive overload across rep ranges — light sets prime the nervous system, moderate sets accumulate volume, and heavy sets drive maximal motor unit recruitment.

Cable Flys: Unlike free-weight flys where tension drops at the top, cable flys maintain constant mechanical tension throughout the full arc of motion. Zeidler et al. (2022) demonstrated that cable-based chest exercises preserve greater shoulder joint moment across both the stretched and contracted positions. This constant tension profile makes cables superior for sustained time-under-tension and metabolic stress — two key hypertrophy drivers.

Dumbbell Fly Stretch: Stretch-mediated hypertrophy is one of the most significant findings in recent exercise science. A meta-analysis of 30 RCTs found that training at long muscle lengths produced significantly greater hypertrophy with effects greatest at long muscle lengths ($d = 0.30$, $p < 0.001$) (Warneke et al., 2024). Inter-set passive stretching may also sensitize stretch-activated signaling pathways, offering a time-efficient hypertrophic stimulus (Schoenfeld, De Souza & Wackerhage, 2022). Light dumbbell flys with a deep stretch at the bottom capitalize on this mechanism, providing a powerful hypertrophic stimulus even at low loads.

THIS IS NOT MEDICAL OR HEALTH ADVICE. FOR RESEARCH & EDUCATIONAL PURPOSES ONLY. CONSULT A PHYSICIAN BEFORE USE.

Skullcrushers: The triceps long head crosses both the elbow and shoulder joints, meaning it is maximally stretched in overhead or inclined positions. Maeo et al. (2022) conducted a 12-week RCT showing overhead elbow extension produced +16% triceps hypertrophy vs. +10% for neutral-position extension — a clinically significant difference attributable to greater stretch on the long head during the overhead position. Skullcrushers replicate this overhead stretch mechanic by placing the elbows above the head at the bottom of each rep, putting the long head under maximal tension at its most lengthened position. This directly applies the stretch-mediated hypertrophy principle for superior triceps growth. The 4 sets of 100 total reps provide high-volume accumulation that prioritizes the long head — the largest of the three triceps heads and the primary driver of upper arm size when viewed from the side.

Tricep Pulldown Trifecta: Different elbow joint angles and grip orientations significantly alter the EMG activity-to-force ratio among the three triceps heads (Akima et al., 2021). By rotating through five cable variations — rope, V-bar, straight bar, reverse grip, and single-arm — the trifecta ensures comprehensive triceps stimulation across all heads. Barbell overhead press data confirms pressing movements produce high triceps excitation across all conditions (Ce, Coratella et al., 2022), supporting multi-angle programming for maximal triceps development. The 10-set total across five grip variations accumulates 150-200 reps of direct triceps work, providing enormous metabolic stress and cellular swelling that complement the mechanical tension from skullcrushers.

◆ PULL DAY — BACK, REAR DELTS & BICEPS ◆

TUESDAY & FRIDAY | PERFORMED 2x PER WEEK

◆ Sometimes going 20% lighter and pause-repping can make you stronger faster. Ego is the enemy of gains.

EXERCISE	SETS	REP TARGET	NOTES
Lat Pulldown Trifecta (Wide / Neutral / Close)	3 (x3 grips)	54	18 reps per grip variation
Cable Rows	3	30	Full contraction, 1s squeeze
Cable Face Pulls	2	100	High-rep posterior delt / rotator cuff
Rear Delt Machine	4	60	15 reps per set, controlled
Dumbbell Rows	3	30	10 per set, heavy and strict
Bicep Curl Trifecta	3	135	Hammers x15 > Oscillating x15 > Side Hammers x15, 30s rest, x3
Bicep Curl Machine Bi-Burn	3	60	20 reps per set, drop set on last
Cable Curls	3	150	50 reps per set — high-volume finisher

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

RESEARCH: WHY THESE EXERCISES

Lat Pulldown Trifecta: Grip width modulates secondary mover activation without significantly altering latissimus dorsi recruitment. Andersen et al. (2014) found shoulder-width grip produced the greatest lat activation, while Lusk et al. (2010) showed supinated grip increased biceps brachii involvement. The trifecta approach (wide, neutral, close) ensures complete lat fiber coverage and varied bicep/forearm stimulation across grips (Signorile, Zink & Szwed, 2002). Recent EMG analysis confirms posterior deltoid and biceps activation varies substantially across grip orientations (Buonsenso et al., 2025), validating the multi-grip strategy.

Cable Rows: Seated cable rows with full scapular retraction produce significantly greater middle trapezius and rhomboid activation compared to pulldown variations (Lehman et al., 2004). The horizontal pull pattern is essential for mid-back thickness and scapular health, complementing the vertical pull of lat pulldowns. Full contraction with a 1-second squeeze ensures peak scapular retractor engagement.

Face Pulls: External rotation at 90 degrees abduction maximizes posterior deltoid and infraspinatus activation (Hawkes et al., 2015). High-rep face pulls serve dual purposes: posterior deltoid hypertrophy and rotator cuff prehabilitation. The 2 sets of 100 total reps accumulate enormous time-under-tension for the external rotators, building the structural integrity needed to support heavy pressing 4x/week. This is a non-negotiable shoulder health exercise that prevents the anterior-dominant imbalances common in pressing-heavy programs.

Rear Delt Machine: Internal rotation of the humerus during lateral-raise-type movements dramatically increases posterior deltoid EMG with effect sizes of 10.79-21.34 compared to other variations (Coratella et al., 2020). Pronated grip produces greater rear delt activation than hammer grip during the eccentric phase (Demirtas et al., 2023). Machine isolation provides consistent resistance curves for targeted rear delt development.

Dumbbell Rows: Unilateral dumbbell rows allow for greater range of motion and independent limb training, addressing bilateral strength asymmetries that bilateral movements mask. The heavy, strict execution (10 reps per set) prioritizes mechanical tension on the latissimus dorsi and lower trapezius, complementing the higher-rep cable work earlier in the session. The unilateral loading also engages the obliques and quadratus lumborum as stabilizers, providing functional core work without dedicated ab training. Keep strict form — eliminating momentum ensures the target muscles bear the full load through the entire range of motion.

Bicep Curl Trifecta & Variations: Supinated grip produces 6-9% greater biceps brachii RMS activation vs. pronated grip (Coratella et al., 2023). EZ bar curls show higher combined biceps and brachioradialis activation in both concentric and eccentric phases (Paoli et al., 2018). Training at long muscle lengths (incline curls) produces greater distal biceps hypertrophy than short-length training like preacher curls (Zubillaga et al., 2023). The multi-grip trifecta cycles through hammer curls, oscillating curls, and side hammer curls to diversify both mechanical and neural stimuli for complete arm development. The high-volume cable curl finisher (50 reps per set) maximizes metabolic stress and cellular swelling, driving sarcoplasmic hypertrophy that complements the myofibrillar gains from heavier curl work. The Bicep Curl Machine bi-burn with drop sets on the last set pushes the biceps past concentric failure, recruiting additional motor units during the eccentric phase for maximal fiber damage and adaptation stimulus.

◆ SHOULDERS, TRAPS & LEGS ◆

WEDNESDAY & SATURDAY | PERFORMED 2x PER WEEK

◆ SHOULDERS & TRAPS

EXERCISE	SETS	REP TARGET	NOTES
Shoulder Burner Heavy	3	10	Heavy lateral raises — strict form
Shoulder Burner Volume	5-6	100-120	Light laterals, high rep burn
Shrugs (3s Pause at Top)	3	10	3-second hold at peak contraction
Shoulder Press (Pyramid)	5	65 total	20 / 15 / 12 / 10 / 8 reps — increasing weight

◆ LEGS

EXERCISE	SETS	REP TARGET	NOTES
Leg Extensions	3	45	15 per set, full contraction squeeze
Leg Press	4	40	10 per set, controlled depth
Hamstring Curls	3	45	15 per set, slow eccentric
Calf Raises	4	100	25 per set, full ROM + pause at stretch

RECOVERY: 15 MIN SAUNA | Hydrate with electrolytes | Target 80-100 C

WEEKLY STRUCTURE

MON & THU = Push | TUE & FRI = Pull | WED & SAT = Shoulders/Legs

SUNDAY = REST & RECOVER

RESEARCH: WHY THESE EXERCISES

Lateral Raises (Shoulder Burner): Neutral humerus rotation during lateral raise maximizes medial deltoid activation at 30.3% MVIC, outperforming frontal raise and all rotation variants (Coratella et al., 2020). Lateral raises produce significantly higher medial deltoid activation than bench press or shoulder press (Campos et al., 2020). Both dumbbell and cable lateral raises yield similar hypertrophy (3.3-4.6% thickness increase) over 8 weeks (Larsen et al., 2025), confirming either modality is effective. The heavy/volume split maximizes both mechanical tension and metabolic stress.

THIS IS NOT MEDICAL OR HEALTH ADVICE. FOR RESEARCH & EDUCATIONAL PURPOSES ONLY. CONSULT A PHYSICIAN BEFORE USE.

Shrugs (3s Pause): Shrugs produce significantly greater upper trapezius EMG activation compared to all tested alternatives (Kim et al., 2016). High-intensity shrug loading results in sustained upper trapezius activation (Sjogaard et al., 2020). The 3-second isometric hold at peak contraction extends time-under-tension for the upper traps beyond what standard concentric/eccentric reps provide, maximizing mechanical stimulus per rep.

Shoulder Press (Pyramid): Front barbell overhead press maximizes anterior deltoid and pectoralis major excitation, while back press enhances medial and posterior deltoid activation. Barbell press uniformly exceeds machine press in overall muscle excitation (Ce, Coratella et al., 2022). Shoulder press also produces significantly higher anterior and medial deltoid activation than bench press and dumbbell fly (Campos et al., 2020). The pyramid rep scheme (20 down to 8) combines endurance, hypertrophy, and strength stimuli in a single exercise.

Leg Extensions: Systematic review confirms leg extension movements produce maximum vastus medialis (VMO) isolation, with foot position and ROM adjustments modulating quad-head-specific activation (Martin-Fuentes, Oliva-Lozano & Muyor, 2020). Full contraction squeeze at the top ensures peak quadriceps activation at the shortened position.

Leg Press: Leg press produces comparable quadriceps activation to the squat but with significantly lower erector spinae and gluteus maximus recruitment (Saeterbakken, Stien & Andersen, 2021). Foot placement modulates quad vs. glute/hamstring involvement — high narrow stance maximizes vastus lateralis activation while low stance increases gluteus maximus contribution (Martin-Fuentes et al., 2020). A quad-dominant compound without axial compression risk.

Hamstring Curls: A 10-week training study confirmed leg curls significantly increase biceps femoris long head muscle thickness and isometric torque regardless of ankle position (Cadeo et al., 2023). Nordic hamstring curl research further supports eccentric hamstring training for injury prevention and fascicle length gains across 39 studies reviewed (Sharma, Desai & Devare, 2024). Slow eccentrics amplify the hypertrophic stimulus.

Calf Raises: Standing calf raise produced 5.6% total triceps surae hypertrophy vs. 2.1% for seated over 12 weeks, driven by greater gastrocnemius activation at long muscle lengths (Kanehisa et al., 2023). Critically, partial ROM performed in the dorsiflexed (stretched) position produced 15.2% medial gastrocnemius hypertrophy vs. only 6.7% for full ROM and 3.4% for shortened position (Kassiano et al., 2023). Full ROM with emphasis on the bottom stretch is optimal.

◆ TRAINING METHODOLOGY — THE SCIENCE ◆

6-DAY PPL FREQUENCY

The Leonidas Lifts program trains each muscle group twice per week using a Push/Pull/Shoulders & Legs rotation across six training days. This frequency is directly supported by Schoenfeld, Ogborn & Krieger's (2016) meta-analysis of 22 studies, which found training each muscle group 2x/week produced significantly greater hypertrophy than 1x/week (effect size 0.37 higher), with a trend for even greater benefit at 3x/week when volume is matched. A subsequent meta-analysis (Schoenfeld, Grgic & Krieger, 2019) clarified that higher frequency primarily benefits hypertrophy by facilitating greater total weekly volume accumulation — which is the true driver of muscle growth. The 6-day structure distributes volume optimally while providing 48-72 hours between sessions for the same muscle group, matching the muscle protein synthesis recovery window.

WEEKLY VOLUME: SETS PER MUSCLE GROUP

This program delivers 18-29 direct sets per muscle group per week — well within the evidence-based optimum. Schoenfeld, Ogborn & Krieger (2017) established a dose-response relationship where 10+ sets per muscle per week produced significantly greater hypertrophy, with a continuing benefit trend up to ~20 sets/week. The Bayesian network meta-analysis by Currier et al. (2023) ranked higher-volume multiset training performed twice weekly as the highest-probability protocol for hypertrophy, with 3+ sets per exercise consistently associated with superior gains. Push day delivers ~29 sets for chest and ~14 for triceps; Pull day delivers ~24 for back and ~12 for biceps; Shoulders & Legs day delivers ~16 for shoulders and ~14 for legs — all exceeding the 10-set threshold.

PROGRESSIVE OVERLOAD

Progressive overload is the single most important training variable. Plotkin et al. (2022) conducted an 8-week RCT comparing load progression vs. repetition progression and found both produced equivalent hypertrophy and strength gains when effort was equated. This confirms that progressive overload — not specifically progressive load — is the critical variable. In this program, the bench press pyramid provides load-based progression, while high-rep exercises (cable curls, face pulls) use rep-based progression. Both pathways drive adaptation when effort is maximized.

Track every session. Beat the logbook.

REP RANGES: THE HYPERTROPHY CONTINUUM

This program intentionally spans rep ranges from 3-5 (heavy bench sets) to 50+ (cable curls). Schoenfeld et al. (2021) re-examined the repetition continuum and confirmed that low (1-5), moderate (6-12), and high (15-30+) rep ranges all produce equivalent hypertrophy when sets are taken to or near muscular failure. A volume-matched meta-analysis (Carvalho et al., 2022) reinforced that hypertrophic adaptations are equivalent across 30-90% 1RM loads. The variety of rep ranges maximizes both mechanical tension (heavy sets) and metabolic stress (high-rep sets) for comprehensive muscle development. Heavy sets additionally drive myofibrillar protein synthesis and strength gains that support future progressive overload capacity.

REST DAY PLACEMENT & RECOVERY WINDOWS

Sunday serves as the sole full rest day, strategically placed after the second Shoulders & Legs session (Saturday). Muscle protein synthesis (MPS) peaks 24-48 hours post-exercise and returns to baseline by 48-72 hours. The PPL rotation ensures each muscle group is re-stimulated within this MPS window: Push on Monday/Thursday, Pull on Tuesday/Friday, and Shoulders & Legs on Wednesday/Saturday. This maximizes the time spent in an elevated MPS state across the week. The Sunday rest day allows systemic recovery — nervous system restoration, glycogen replenishment, hormonal rebalancing, and psychological recovery — before the next training cycle begins. For athletes in a caloric deficit, this rest day is especially critical as recovery capacity is compromised.

INTENSITY TECHNIQUES: WHEN TO USE THEM

This program incorporates several intensity techniques beyond straight sets. **Drop sets** (Bicep Curl Machine) extend the set beyond initial failure by reducing load, accumulating additional metabolic stress. **Trifecta supersets** (Lat Pulldown, Bicep Curl, Tricep Pulldown) cycle through multiple grip positions with minimal rest, maximizing time-under-tension across different joint angles. **Isometric holds** (Shrug 3s pause) increase time-under-tension at peak contraction. **High-rep finishers** (Cable Curls 50 reps) drive metabolic stress and cellular swelling. These techniques should be applied as programmed — adding extra intensity techniques to already high-volume sessions risks exceeding recovery capacity, particularly during caloric deficits.

◆ RECOVERY SCIENCE ◆

RESEARCH-BACKED PROTOCOLS FOR NATURAL ATHLETES

1. SAUNA PROTOCOL

Post-workout sauna at 80-100 C activates heat shock proteins (HSP70/HSP90) that prevent muscle protein degradation and assist satellite cell repair. These molecular chaperones stabilize damaged proteins and facilitate proper refolding — essential for recovery from the high-volume training in this program.

A single 15-minute session drives HSP70/HSP72 elevation; a series of 10 Finnish sauna baths (3 x 15 min each) modulates the immune response and HSP70 family levels in both trained and untrained men (Czerwinska-Ledwig et al., 2023). The landmark Finnish study documented significant GH increases with repeated sauna exposure at 80 C — though the peak response requires longer sessions; a 15-minute post-workout protocol yields a meaningful acute GH response that stacks with exercise-induced GH release (Leppaluoto et al., 1986).

Infrared sauna reduces DOMS and preserves explosive neuromuscular performance vs. passive rest. A single 20-minute infrared session at 43 C after resistance exercise improved recovery metrics in male athletes (Ahokas et al., 2022). **In a caloric deficit:** consume 500-750 mL electrolyte fluid before/during the session and rehydrate to body weight parity. Sweat losses of 0.5-1.0 kg per session deplete sodium, potassium, and magnesium — supplement accordingly. Target 80-100 C for Finnish sauna or 43-55 C for infrared.

2. SLEEP & GH RELEASE

Approximately 70-75% of total daily GH is secreted during nocturnal sleep, with the first slow-wave sleep (SWS) episode accounting for 50-60% of the nocturnal total (Takahashi, Kipnis & Daughaday, 1968). Growth hormone is the primary anabolic hormone released during sleep — it drives muscle protein synthesis, fat mobilization, and connective tissue repair. Disrupting the first SWS cycle has a disproportionate impact on GH output.

A single night of sleep deprivation drops muscle protein synthesis by ~18%, increases cortisol ~21%, and decreases testosterone ~22% (Lamon et al., 2020). On a 6-day high-volume split, **7.5-8 hours minimum** ensures 4-5 full sleep cycles and sufficient SWS for peak GH release. Cutting to 6 hours reduces total SWS by 20-30%, directly compromising recovery capacity and hormonal optimization.

Magnesium supplementation (300-400 mg, 30-60 min before bed) improves sleep efficiency, reduces cortisol, and increases melatonin. Magnesium acts as an NMDA antagonist and GABA-A agonist, promoting neurological relaxation. An 8-week RCT in 46 subjects showed 500 mg/day magnesium significantly improved ISI score, sleep efficiency, and early morning awakening (Abbasi et al., 2012). Pair with a consistent sleep schedule — irregular bedtimes fragment SWS architecture even when total hours are adequate.

3. DELOAD PROTOCOLS

On a 6-day split, **deload every 4-5 weeks** during a caloric deficit (6-8 weeks in a surplus). An international Delphi consensus panel reached 100% agreement that deloads must be included in each mesocycle, with timing responsive to athlete fatigue markers (Strafford et al., 2023). The consensus defines deloading as "a period of reduced training stress designed to mitigate physiological and psychological fatigue, promote recovery, and enhance preparedness for subsequent training." Volume reduction of 30-50% with maintained or reduced intensity was the consensus recommendation.

Active recovery (maintain frequency, reduce volume 50-60% and intensity 30-40%) is marginally superior to complete rest — both preserve strength, but active deloads show slightly higher molecular remodeling markers including MHC-IIa mRNA expression (Vann et al., 2021). During deficit deloads, consider raising calories to maintenance temporarily to restore anabolic signaling without compromising fat loss trajectory (Pinero & Schoenfeld, 2024). A well-timed deload re-sensitizes the muscle to the training stimulus, allowing the next mesocycle to drive new adaptation.

4. CONNECTIVE TISSUE

Tendons adapt far slower than muscle (collagen half-life: 60-100 years vs. muscle protein turnover in days). Joint warning signs are **leading indicators** of overtraining — pain, clicking, or stiffness that persists beyond warmup signals accumulated connective tissue strain. On a 6-day high-volume split, proactive joint care is non-negotiable.

UC-II Collagen 40 mg/day works via oral tolerance — it activates regulatory T-cells that suppress joint inflammation. An RCT in healthy volunteers showed the UC-II group could exercise 2x longer before knee pain onset (2.8 vs. 1.4 min, $p < 0.05$) and showed significantly improved knee extension at 120 days (Lugo et al., 2013). **Aquamin** (2400 mg/day seaweed-derived minerals) significantly improved knee pain, stiffness, and walking distance (+101 feet, +7% vs baseline) in OA patients —

superior to glucosamine sulfate and placebo (Frestedt et al., 2008).

BPC-157 & TB-500: Robust preclinical data for tendon healing and angiogenesis, but **no published human RCTs** exist as of 2026. Both are WADA-banned and constitute unregulated self-experimentation. An injectable peptide review confirms BPC-157 shows potential for tendon/muscle/bone repair in animal models, while TB-500 (thymosin beta-4 fragment) promotes angiogenesis, but current evidence is insufficient to support clinical recommendation (Mayfield et al., 2026). The evidence-based position: use proven interventions (UC-II, Aquamin, vitamin C, load management) while monitoring for eventual human trial data.



“Seventeen years under the bar before I ever picked up a syringe. Earn the foundation first. Then build on it.”

· **LEONIDASLIFTS.COM**



NO STEROIDS. JUST WORK.

◆ REFERENCES ◆

- Abbasi, B., Kimiagar, M., Sadeghnia, K., Shirazi, M. M., Hedayati, M., & Rashidkhani, B. (2012). The effect of magnesium supplementation on primary insomnia in elderly: A double-blind placebo-controlled clinical trial. *Journal of Research in Medical Sciences*. <https://pubmed.ncbi.nlm.nih.gov/23853635/>
- Ahokas, E. K., Savolainen, E., Kyrolainen, H., Hanstock, H. G., & Ihalainen, J. K. (2022). A post-exercise infrared sauna session improves recovery of neuromuscular performance and muscle soreness after resistance exercise training. *Biology of Sport*. <https://doi.org/10.5114/biolsport.2023.119289>
- Akima, H., Ushiyama, J., Enomoto, Y., Ogawa, M., & Takahashi, H. (2021). Effect of elbow joint angles on EMG activity versus force relationships of synergistic muscles of the triceps brachii. *PLOS ONE*. <https://doi.org/10.1371/journal.pone.0252644>
- Andersen, V., Fimland, M. S., Wiik, E., Skoglund, A., & Saeterbakken, A. H. (2014). Effects of grip width on muscle strength and activation in the lat pull-down. *J Strength Cond Res*. <https://doi.org/10.1097/JSC.0000000000000232>
- Buonsenso, A., Fiorilli, G., Gentile, A., et al. (2025). EMG analysis of back muscle activation during lat pulldown exercise: Effects of grip variations. *J Functional Morphology and Kinesiology*. <https://doi.org/10.3390/jfmk10030345>
- Cabral, H. V., Resende, R. A., et al. (2021). Non-uniform excitation of the pectoralis major muscle during flat and inclined bench press. *Scand J Med Sci Sports*. <https://doi.org/10.1111/sms.14082>
- Cadeo, G. M., Fujita, R. A., et al. (2023). Myoelectric activity and improvements in strength and hypertrophy are unaffected by ankle position during prone leg curl. *Eur J Sport Sci*. <https://doi.org/10.1080/17461391.2023.2214794>
- Campos, Y., Vianna, J., Guimaraes, M., et al. (2020). Different shoulder exercises affect the activation of deltoid portions. *J Human Kinetics*. <https://doi.org/10.2478/hukin-2020-0033>
- Carvalho, L., Junior, R. M., Barreira, J., Schoenfeld, B. J., et al. (2022). Muscle hypertrophy and strength gains after resistance training with different volume-matched loads. *Appl Physiol Nutr Metab*. <https://doi.org/10.1139/apnm-2021-0515>
- Ce, E., Coratella, G., Tornatore, G., Longo, S., & Esposito, F. (2020). An EMG analysis of lateral raise variations and frontal raise in competitive bodybuilders. *Int J Environ Res Public Health*. <https://doi.org/10.3390/ijerph17176015>
- Ce, E., Coratella, G., Tornatore, G., Longo, S., & Esposito, F. (2022). Front vs back and barbell vs machine overhead press. *Frontiers in Physiology*. <https://doi.org/10.3389/fphys.2022.825880>
- Ce, E., Coratella, G., Tornatore, G., et al. (2023). Biceps brachii and brachioradialis excitation in biceps curl exercise: Different handgrips, different synergy. *Sports*. <https://doi.org/10.3390/sports11030064>
- Czerwinska-Ledwig, O., Pilch, W., Szarek, M., et al. (2023). Effects of Finnish sauna sessions on immune response and HSP-70 levels. *Int J Hyperthermia*. <https://doi.org/10.1080/02656736.2023.2179672>
- Currier, B. S., McLeod, J. C., Banfield, L., et al. (2023). Resistance training prescription for muscle strength and hypertrophy in healthy adults. *Br J Sports Med*. <https://doi.org/10.1136/bjsports-2023-106807>
- Demirtas, B., Cakir, O., Cetin, O., & Cilli, M. (2023). The effects of handgrip and ROM variations on muscle activity in different deltoid exercises. *Kinesiology Slovenica*. <https://doi.org/10.52165/kinsi.29.1.73-87>
- Frestedt, J. L., Walsh, M., Kuskowski, M. A., & Zenk, J. L. (2008). A natural mineral supplement provides relief from knee osteoarthritis symptoms. *Nutrition Journal*. <https://doi.org/10.1186/1475-2891-7-9>
- Hawkes, D. H., Kemp, G. J., Alizadehkhayat, O., & Frostick, S. P. (2015). EMG analysis of shoulder girdle musculature during external rotation exercises. *Orthop J Sports Med*. <https://doi.org/10.1177/2325967115613988>
- Kanehisa, H., Yata, H., Ikegawa, S., & Fukunaga, T. (2023). Triceps surae muscle hypertrophy is greater after standing versus seated calf-raise training. *Frontiers in Physiology*. <https://doi.org/10.3389/fphys.2023.1272106>
- Kassiano, W., Costa, B., Nunes, J. P., et al. (2023). Greater gastrocnemius muscle hypertrophy after partial ROM training at long muscle lengths. *J Strength Cond Res*. <https://doi.org/10.1519/JSC.0000000000004460>
- Kim, M. J., Jeong, S. M., Park, S. K., & Park, D. J. (2016). Intramuscular activation of scapular stabilizing muscles during shrug exercise. *J Korean PNF Association*. <https://doi.org/10.21598/JKPNFA.2016.14.2.113>
- Lamon, S., Morabito, A., Arentson-Lantz, E. J., et al. (2020). The effect of acute sleep deprivation on skeletal muscle protein synthesis and the hormonal environment. *Physiological Reports*. <https://doi.org/10.14814/phy2.14660>
- Larsen, S., Kristiansen, E., & van den Tillaar, R. (2025). Dumbbell versus cable lateral raises for lateral deltoid hypertrophy. *Frontiers in Physiology*. <https://doi.org/10.3389/fphys.2025.1611468>
- Lehman, G. J., Buchan, D. D., Lundy, A., Myers, N., & Nalborczyk, A. (2004). Variations in muscle activation levels during traditional latissimus dorsi weight training exercises. *Chiropractic & Osteopathy*. <https://doi.org/10.1186/1476-5918-3-4>

- Leppaluoto, J., Huttunen, P., Hirvonen, J., et al. (1986). Endocrine effects of repeated sauna bathing. *Acta Physiologica Scandinavica*. <https://doi.org/10.1111/j.1748-1716.1986.tb08000.x>
- Lugo, J. P., Saiyed, Z. M., Lau, F. C., et al. (2013). UC-II for joint support: a randomized, double-blind, placebo-controlled study in healthy volunteers. *JISSN*. <https://doi.org/10.1186/1550-2783-10-48>
- Lusk, S. J., Hale, B. D., & Russell, D. M. (2010). Grip width and forearm orientation effects on muscle activity during the lat pull-down. *J Strength Cond Res*. <https://doi.org/10.1519/JSC.0b013e3181ddb0ab>
- Maeo, S., Wu, Y., Huang, M., et al. (2022). Triceps brachii hypertrophy is substantially greater after elbow extension training performed in the overhead versus neutral arm position. *Eur J Sport Sci*. <https://doi.org/10.1080/17461391.2022.2100279>
- Martin-Fuentes, I., Oliva-Lozano, J. M., & Muyor, J. M. (2020). Evaluation of the lower limb muscles' EMG activity during the leg press exercise: A systematic review. *Int J Environ Res Public Health*. <https://doi.org/10.3390/ijerph17134626>
- Mayfield, C. K., Bolia, I. K., Feingold, C. L., et al. (2026). Injectable peptide therapy: A primer for orthopaedic and sports medicine physicians. *Am J Sports Med*. <https://doi.org/10.1177/03635465251357593>
- Paoli, A., Marcolin, G., Petrone, N., et al. (2018). Differences in EMG activity of biceps brachii and brachioradialis while performing three variants of curl. *PeerJ*. <https://doi.org/10.7717/peerj.5165>
- Pinero, A., Schoenfeld, B. J., Oberlin, D., et al. (2024). Gaining more from doing less? The effects of a one-week deload period during supervised resistance training on muscular adaptations. *PeerJ*. <https://doi.org/10.7717/peerj.16777>
- Plotkin, D. L., Coleman, M., Van Every, D. W., et al. (2022). Progressive overload without progressing load? The effects of load or repetition progression on muscular adaptations. *PeerJ*. <https://doi.org/10.7717/peerj.14142>
- Rodriguez-Ridao, D., Antequera-Vique, J. A., Martin-Fuentes, I., & Muyor, J. M. (2020). Effect of five bench inclinations on the EMG activity of the pectoralis major, anterior deltoid, and triceps brachii. *Int J Environ Res Public Health*. <https://doi.org/10.3390/ijerph17197339>
- Saeterbakken, A. H., Stien, N., & Andersen, V. (2021). EMG comparison of five lower-limb muscles between single- and multi-joint exercises. *J Sports Sci Med*. <https://doi.org/10.52082/jssm.2021.56>
- Schoenfeld, B. J., De Souza, E. O., & Wackerhage, H. (2022). Inter-set stretch: A potential time-efficient strategy for enhancing skeletal muscle adaptations. *Front Sports Active Living*. <https://doi.org/10.3389/fspor.2022.1035190>
- Schoenfeld, B. J., Grgic, J., & Krieger, J. (2019). How many times per week should a muscle be trained to maximize hypertrophy? *J Sports Sci*. <https://doi.org/10.1080/02640414.2018.1555906>
- Schoenfeld, B. J., Grgic, J., Van Every, D. W., & Plotkin, D. L. (2021). Loading recommendations for muscle strength, hypertrophy, and local endurance: A re-examination of the repetition continuum. *Sports*. <https://doi.org/10.3390/sports9020032>
- Schoenfeld, B. J., Ogborn, D., & Krieger, J. W. (2016). Effects of resistance training frequency on measures of muscle hypertrophy: A systematic review and meta-analysis. *Sports Med*. <https://doi.org/10.1007/s40279-016-0543-8>
- Schoenfeld, B. J., Ogborn, D., & Krieger, J. W. (2017). Dose-response relationship between weekly resistance training volume and increases in muscle mass. *J Sports Sci*. <https://doi.org/10.1080/02640414.2016.1210197>
- Sharma, V., Desai, S., & Devare, N. (2024). The role of the Nordic hamstring curl in the rehabilitation of hamstring injuries: A narrative review. *Parul Univ J Health Sci Res*. <https://doi.org/10.62373/pujhsr.2023.123>
- Signorile, J. F., Zink, A. J., & Szwed, S. P. (2002). A comparative EMG investigation of muscle utilization patterns using various hand positions during the lat pull-down. *J Strength Cond Res*. [doi:10.1519/1533-4287\(2002\)016](https://doi.org/10.1519/1533-4287(2002)016)
- Sjogaard, G., Sogaard, K., Kristensen, L. A., et al. (2020). EMG evaluation of specific elastic band exercises targeting neck and shoulder muscle activation. *Applied Sciences*. <https://doi.org/10.3390/app10030756>
- Strafford, B. W., Nolan, D., Bell, L., et al. (2023). Integrating deloading into strength and physique sports training programmes: An international Delphi consensus approach. *Sports Med Open*. <https://doi.org/10.1186/s40798-023-00633-0>
- Takahashi, Y., Kipnis, D. M., & Daughaday, W. H. (1968). Growth hormone secretion during sleep. *J Clinical Investigation*. <https://doi.org/10.1172/JCI105893>
- Vann, C. G., Haun, C. T., Osburn, S. C., et al. (2021). Molecular differences in skeletal muscle after 1 week of active vs. passive recovery from high-volume resistance training. *J Strength Cond Res*. <https://doi.org/10.1519/JSC.0000000000004071>
- Warneke, K., Lohmann, L. H., Lima, C. D., et al. (2024). Effects of chronic static stretching on maximal strength and muscle hypertrophy: A systematic review and meta-analysis. *Sports Med Open*. <https://doi.org/10.1186/s40798-024-00706-8>
- Zeidler, F., Mileva, K. N., et al. (2022). Chest exercises: Movement and loading of shoulder, elbow and wrist joints. *Sports*. <https://doi.org/10.3390/sports10020019>
- Zubillaga, A., Zabaleta-Korta, A., Fernandez-Pena, E., et al. (2023). Regional hypertrophy: The effect of exercises at long and short muscle lengths in recreationally trained women. *J Human Kinetics*. <https://doi.org/10.5114/jhk/163561>

- Harty, P. S., Zabriskie, H. A., Stecker, R. A., Currier, B. S., Moon, J. M., Jagim, A. R., & Kerksick, C. M. (2019). Upper and lower thresholds of fat-free mass index in a large cohort of female collegiate athletes. *Journal of Sports Sciences*, 37(20), 2381–2388. <https://doi.org/10.1080/02640414.2019.1634964>
- Jagim, A. R., Fields, J. B., Jones, M. T., Byrd, M., & Kerksick, C. M. (2019). Fat-free mass index in a diverse sample of male collegiate athletes. *Journal of Strength and Conditioning Research*, 33(6), 1474–1479. <https://doi.org/10.1519/JSC.0000000000003154>
- Jagim, A. R., Harty, P. S., Jones, M. T., Fields, J. B., Magee, M., Smith-Ryan, A. E., Luedke, J., & Kerksick, C. M. (2024). Fat-free mass index in sport: Normative profiles and applications for collegiate athletes. *Journal of Strength and Conditioning Research*, 38(9), 1687–1693. <https://doi.org/10.1519/JSC.0000000000004864>
- Kouri, E. M., Pope, H. G., Jr., Katz, D. L., & Oliva, P. (1995). Fat-free mass index in users and nonusers of anabolic-androgenic steroids. *Clinical Journal of Sport Medicine*, 5(4), 223–228. <https://doi.org/10.1097/00042752-199510000-00003>
- Kudsk, K. A., Munoz-Del-Rio, A., Busch, R. A., Kight, C. E., & Schoeller, D. A. (2017). Stratification of fat-free mass index percentiles for body composition based on National Health and Nutrition Examination Survey III bioelectric impedance data. *JPEN: Journal of Parenteral and Enteral Nutrition*, 41(2), 249–257. <https://doi.org/10.1177/0148607115592672>
- Kyle, U. G., Schutz, Y., Dupertuis, Y. M., & Pichard, C. (2003). Body composition interpretation: Contributions of the fat-free mass index and the body fat mass index. *Nutrition*, 19(7–8), 597–604. [https://doi.org/10.1016/S0899-9007\(03\)00061-3](https://doi.org/10.1016/S0899-9007(03)00061-3)
- Schutz, Y., Kyle, U. U. G., & Pichard, C. (2002). Fat-free mass index and fat mass index percentiles in Caucasians aged 18–98 y. *International Journal of Obesity and Related Metabolic Disorders*, 26(7), 953–960. <https://doi.org/10.1038/sj.ijo.0802037>
- Trexler, E. T., Smith-Ryan, A. E., Blue, M. N. M., Schumacher, R. M., Mayhew, J. L., Mann, J. B., Ivey, P. A., Hirsch, K. R., & Mock, M. G. (2017). Fat-free mass index in NCAA Division I and II collegiate American football players. *Journal of Strength and Conditioning Research*, 31(10), 2719–2727. <https://doi.org/10.1519/JSC.0000000000001737>